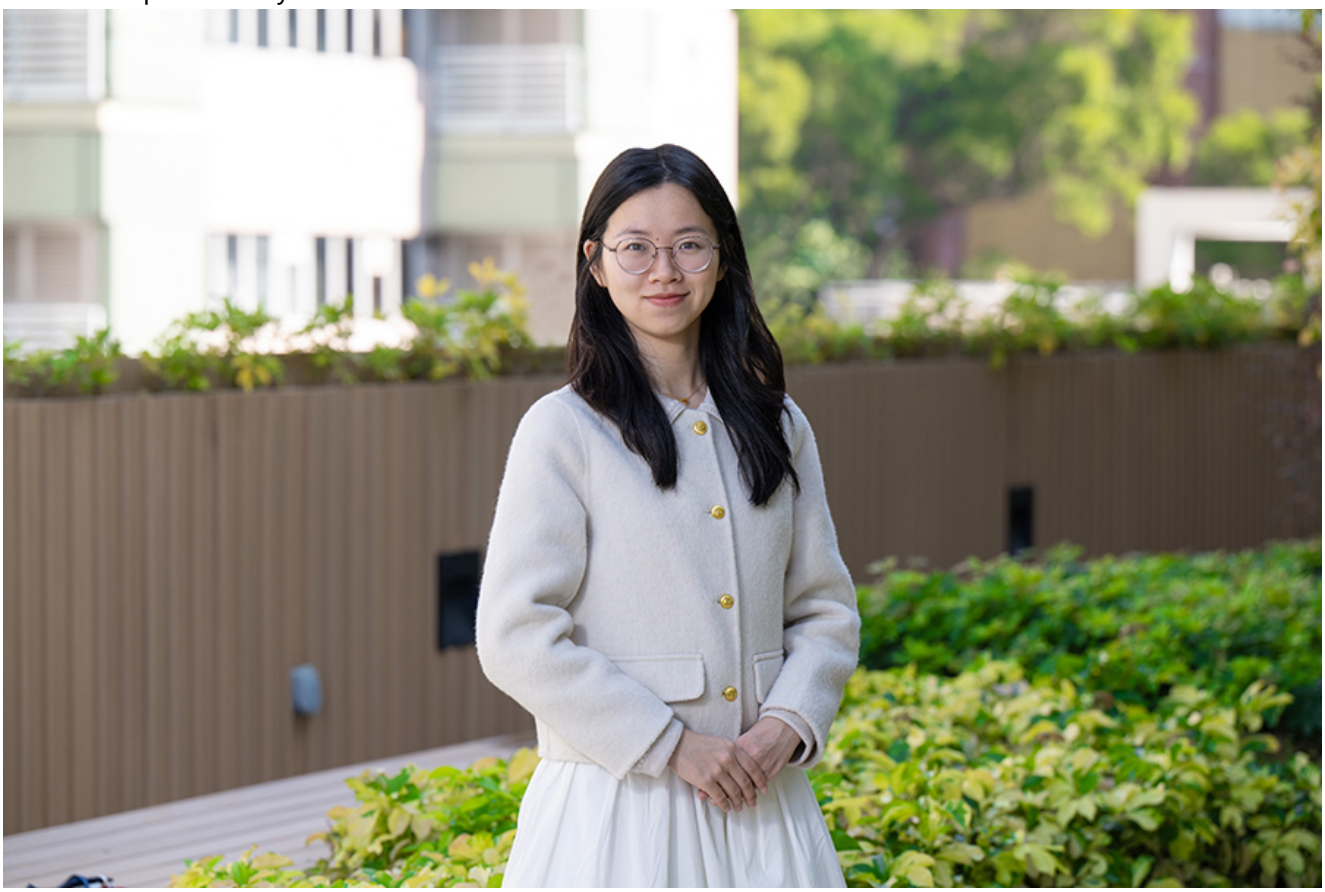

# Prof. GUO, Xiaoqing

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# About

Dr. Guo is an assistant professor in the Department of Computer Science at Hong Kong Baptist University, where she leads **UltraVision+ Lab**. She was a postdoctoral researcher (2023-2024) and visiting researcher (2024-2025) affiliated with the Noble Group in the Department of Engineering Science at the University of Oxford, working on the projects VisualAI and Turing AI WLR Fellowship. She obtained her PhD degree in the Department of Electrical Engineering at City University of Hong Kong in 2022 and received her BEng degree from Beihang University in 2018. She was a research assistant in the Department of Electrical Engineering at Tsinghua University during 2017-2018. In AI and medical imaging fields, she has published over 40 top journal and conference papers, e.g., Nature Biomedical Engineering, TPAMI, IJCV, CVPR, ICCV, ECCV, ICLR, TMI, MedIA, MICCAI, IROS, npj Digital Medicine, and Nature. She has been selected as one of World's Top 80 Chinese Young Female Scholars in AI, the Rising Star of Women in Engineering by Asian Deans' Forum 2024, and she has received prestigious awards, such as CVPR Outstanding Reviewer Award, MSRA Fellowship Nomination Award, Outstanding Doctoral Research Award, Outstanding Research Thesis Award, and three year consecutive Outstanding Academic Performance Award.

* I am looking for PhD students and RA. Please refer to open positions in our lab.

# Research Interests

- AI in Healthcare
- AI for Ultrasound Imaging
- Multi-Modal Learning
- Human-AI Interaction/Collaboration
- Trustworthy Clinical Decision-Making

# Selected Publications

- **Xiaoqing Guo †**, Mohammad Alsharid, He Zhao, Yipei Wang, Jayne Lander, Aris T. Papageorghiou, Alison Noble. "Sonomate: Visually grounded language model for fetal ultrasound understanding and human interaction". **Nature Biomedical Engineering**, 2026.
- **Xiaoqing Guo †**, Qianhui Men, Alison Noble. "MMSummary: Multimodal Summary Generation for Fetal Ultrasound Video". The 27nd International Conference on Medical Image Computing and Computer Assisted Intervention (**MICCAI 2024**), Marrakesh, Morocco.
- **Xiaoqing Guo**, Jie Liu, Yixuan Yuan. "Adaptive Infinite Prototypes for Single Domain Generalization in Semantic Segmentation." International Journal of Computer Vision. 2024. (**IJCV**)
- **Xiaoqing Guo**, Jie Liu, Tongliang Liu, Yixuan Yuan. "Handling Open-set Noise and Novel Target Recognition in Domain Adaptive Semantic Segmentation." IEEE Transactions on Pattern Analysis and Machine Intelligence,

45(8): 9846-9861, 2023. (**TPAMI**)

- **Xiaoqing Guo**, Jie Liu, Tongliang Liu, Yixuan Yuan. “SimT: Handling Open-set Noise for Domain Adaptive Semantic Segmentation.” IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR 2022**), New Orleans, USA.
- **Xiaoqing Guo**, Yixuan Yuan. “Joint Class-Affinity Loss Correction for Robust Medical Image Segmentation with Noisy Labels.” The 25nd International Conference on Medical Image Computing and Computer Assisted Intervention (**MICCAI 2022**), Singapore.
- **Xiaoqing Guo**, Zhen Chen, Jun Liu, Yixuan Yuan. “Non-Equivalent Images and Pixels: Confidence-Aware Resampling with Meta-Learning Mixup for Polyp Segmentation.” Medical Image Analysis, vol. 78, p. 102394, 2022. (**MedIA**)
- **Xiaoqing Guo**, Jie Liu, Yixuan Yuan. “Semantic-oriented Labeled-to-unlabeled Distribution Translation for Image Segmentation.” IEEE Transactions on Medical Imaging, vol. 41, no. 2, pp. 434-445, 2022. (**TMI**)
- **Xiaoqing Guo***, Chen Yang*, Baopu Li, Yixuan Yuan. “MetaCorrection: Domain-aware Meta Loss Correction for Unsupervised Domain Adaptation in Semantic Segmentation.” IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR 2021**).
- **Xiaoqing Guo**, Chen Yang, Yajie Liu, Yixuan Yuan. “Learn to Threshold: ThresholdNet with Confidence-Guided Manifold Mixup for Polyp Segmentation.” IEEE Transactions on Medical Imaging, vol. 40, no. 4, pp. 1134-1146, 2021. (**TMI**)

* indicates equal contribution. † indicates corresponding author.